A Comparative Study of Single-Domain and Multi-Factor Models in Predicting Depression Risk——Taking the RHMCD-20 Dataset as an Example

D760335765
Big Data Analytics in Healthcare

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Background:

Against the backdrop of the COVID-19 pandemic, prolonged isolation and social distancing have exerted an unprecedented impact on mental health, leading to a significant global increase in depression and anxiety symptoms. According to empirical research published in *The Lancet* (2021), the prevalence of major depressive disorder rose sharply in 2020, with females and young adults aged 20–25 being the most severely affected (see Figure 1). This underscores the profound psychological impact of extreme societal shifts on specific demographic structures.

However, traditional screening often relies on single indicators, making it difficult to capture the complex dynamics of mental health. Therefore, this study utilizes the RHMCD-20 dataset (Salehin & Amin, 2024), a multi-faceted survey specifically designed to investigate the quarantine period in Bangladesh. By leveraging machine learning to analyze these multi-dimensional features—including diet, sleep, social weakness, and family history—we aim to overcome the limitations of single-domain models, helping medical professionals identify depression risks more accurately and provide timely support to those in need.

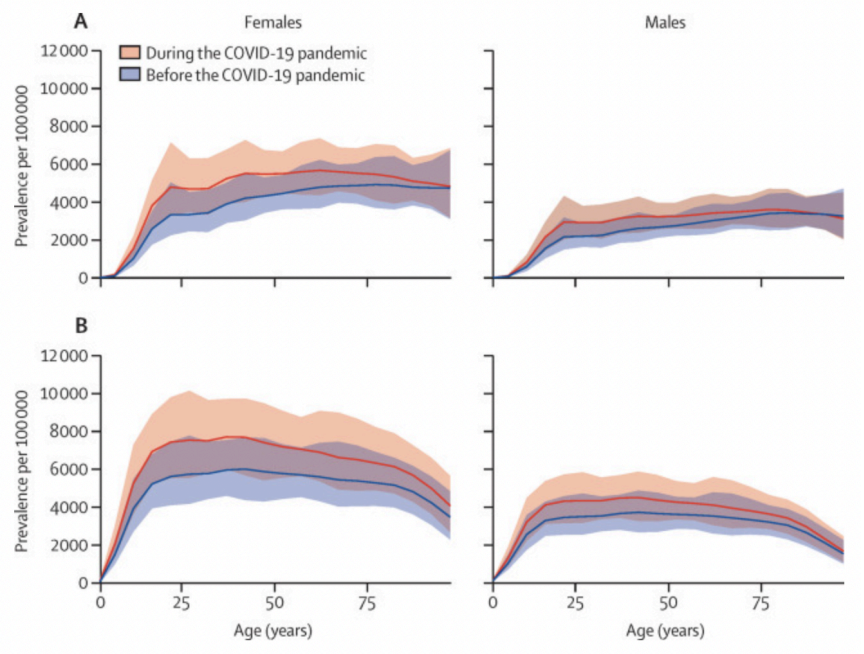

Figure 1: Global prevalence of major depressive disorder (A) and anxiety disorders (B) before and during the COVID-19 pandemic (Source: *The Lancet*, 2021)

Research Question(RQ):

Does a multi-factor model combining lifestyle, psychological, and social variables significantly improve depression risk prediction compared to single-domain models, on the RHMCD-20 dataset?

Hypothesis:

We consider that multi-factor models can significantly improve the predictive ability of depression risk compared to single-feature models.

Expected Contribution:

The expected contribution of this study is to provide a clear empirical basis for demonstrating whether multifactor integration can indeed improve the predictive validity of

depression risk and to assist medical and public health professionals in developing more accurate and actionable early screening strategies.

Group Member:

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Division of Labor Table:

鄭皓仁:Writing a Proposal, Build the Model(Decision tree, Random Forest)

Have good research writing and model construction ability, responsible for proposal writing and building decision tree and random forest model, as the core analyst of the tree model

周楷翔:Writing a Proposal, Data Preprocessing, Build the Model(Naive Bayes, Bernoulli / Multinomial Naive Bayes)

Have experience in data preprocessing and probability modeling. In addition to participating in proposal writing, also responsible for data cleaning and Naive Bayes (Bernoulli/Multinomial) model construction.

許為舜:Produce Presentation PDF, Data Preprocessing

Good at presentation collation and data processing, responsible for data pre-processing, and integrate the analysis results into a presentation PDF to improve the integrity of the presentation.

阮聖倫:Produce Presentation File, Build the Model(XGBoost, KNN)

Focus on model implementation and result display, be responsible for building XGBoost and KNN models, and make final presentation documents to ensure that the model results are clearly conveyed.

施冠宏:Writing a Proposal, Build the Model(SVM, NN)

Combine theoretical understanding and practical modeling ability, participate in proposal writing, and be responsible for the construction of SVM and neural network models, and strengthen the analysis of nonlinear and black box models.

陳彥嘉:Writing a Proposal, Build the Model(Logistic Regression, Bayesian Logistic Regression)

Have statistical and inference ability, participate in proposal writing, and establish Logistic Regression and Bayesian Logistic Regression to provide a strong interpretable benchmark model.

賴亭妤:Video editing, Data Preprocessing

Good at multimedia processing and data collation, responsible for data pre-processing support and video editing of results, so as to make the research results more complete and readable.

Timeline Snapshot:

12/13~12/19: Writing the research proposal

12/20~12/27: Data Preprocessing
12/28~1/9: Modeling
1/10~1/15: Presentation preparation and recording of the final report

Preliminary Literature Review:

1. <<Machine learning models predict the emergence of depression in Argentinean college students during periods of COVID-19 quarantine>>

   The COVID-19 pandemic significantly heightened mental health challenges, particularly depression, among college students, necessitating early risk detection. This study aimed to develop and compare machine learning (ML) models for predicting the emergence of depression in Argentinean college students during the mandatory COVID-19 quarantine. A longitudinal dataset (N=1492) was analyzed using classification (LogReg, RF, SVM) and regression (RR, RF regressor, SVR) models. Features included baseline depression and anxiety scores (T1), history of mental disorder/suicidal behavior, and demographics. The highest predictive performance was achieved by SVM and LogReg classifiers (AUPRC: 0.76) and SVR and RR regressors (R2: 0.56). Univariate models based solely on T1 depression or anxiety scores yielded results comparable to the multivariate models, underscoring the strong predictive relevance of pre-existing conditions. In conclusion, ML models, especially SVM/SVR and LogReg/RR, show promise for the timely detection of at-risk students. Further validation is required before clinical use.

2. <<A machine-learning approach to model risk and protective factors of vulnerability to depression>>

   This study applied machine learning (ML) models to identify key psychosocial risk and protective factors associated with vulnerability to depression for precision interventions. Behavioral data from 496 college-educated participants were analyzed. The Random Forest algorithm was employed for feature selection, revealing four key predictors: three risk factors (brooding, perceived loneliness, and perceived stress) and one protective factor (resilience). A classification model (SVM) built using only these four factors achieved a robust ROC-AUC score of 75.50% in differentiating high- and low-risk groups, a performance comparable to a model using all available factors (ROC-AUC = 77.83%). Based on these selected variables, the researchers developed a practical "mood vulnerability index". These findings provide significant clinical insights for developing efficient and rapid screening tools for individuals at risk of mood disorders and suggest targeted areas for depression intervention programs.

Data Preparation Strategy:

● Missing Data Imputation:

1. For numerical variables, missing values will be imputed using the mean.
2. For categorical variables, missing values will be imputed using the mode.
3. If a variable has a high proportion of missing values (>50%), it may be removed or excluded.

● Feature Engineering:

1. Numerical variables will be standardized using z-score.
2. Categorical variables will be transformed using one-hot encoding.

3. New combined features may be created (eg. For example, a new feature can be created by summing or taking a weighted average of the psychological and lifestyle scores.).
4. Continuous variables, such as age, can be binned into categories (e.g., 19–30, 31–40, 41–50) to facilitate modeling.

Modeling Plan:

1. Linear Models

- Logistic Regression:
  Suitable for predicting binary outcomes, such as whether a person is at risk of depression. It is simple, interpretable, and effective for datasets with moderate features.
- Bayesian Logistic Regression:
  Useful when we want to quantify uncertainty in model parameters or incorporate prior knowledge into prediction.

2. Probabilistic Models:

- Naive Bayes:
  Good for high-dimensional datasets, and can quickly estimate probabilities of depression risk.
- Bernoulli / Multinomial Naive Bayes:
  Suitable for categorical or count-based features, such as survey responses or lifestyle frequency data.

3. Tree-Based Models:

- Decision Tree (CART):
  Easy to interpret and visualize, suitable for initial exploration and understanding feature importance.
- Random Forest:
  Good for datasets with many features and interactions; reduces overfitting compared to a single tree.
- XGBoost:
  Highly accurate and efficient; commonly used in healthcare prediction tasks.

4. Black Box Models:

- K-Nearest Neighbors (KNN):
  A simple and intuitive model that classifies a new sample based on the labels of its nearest neighbors. Useful for predicting depression risk by comparing individuals to others with similar lifestyle, psychological, and social characteristics.
- Support Vector Machine (SVM):
  Application: Predicting depression risk based on multiple features; classifying individuals into risk categories; suitable for high-dimensional data and complex feature interactions.
- Neural Network (NN / Feedforward / MLP):
  A flexible model that can capture complex, non-linear relationships among features. Useful for predicting depression risk based on multiple lifestyle, psychological, and social variables, especially when feature interactions are important.